Agent Skills for full-stack development

Engineering Agentic Workflows: A Full-Stack Perspective

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Slides + code

🤹 Agent Skills for full-stack development

Outline

- **What's an agent skills**
- Skills for full-stack development
- Making a skill and giving it memory
- Token observability

What’s an agent skill

Before we answer that, let’s answer what’s an MCP.

What's an MCP

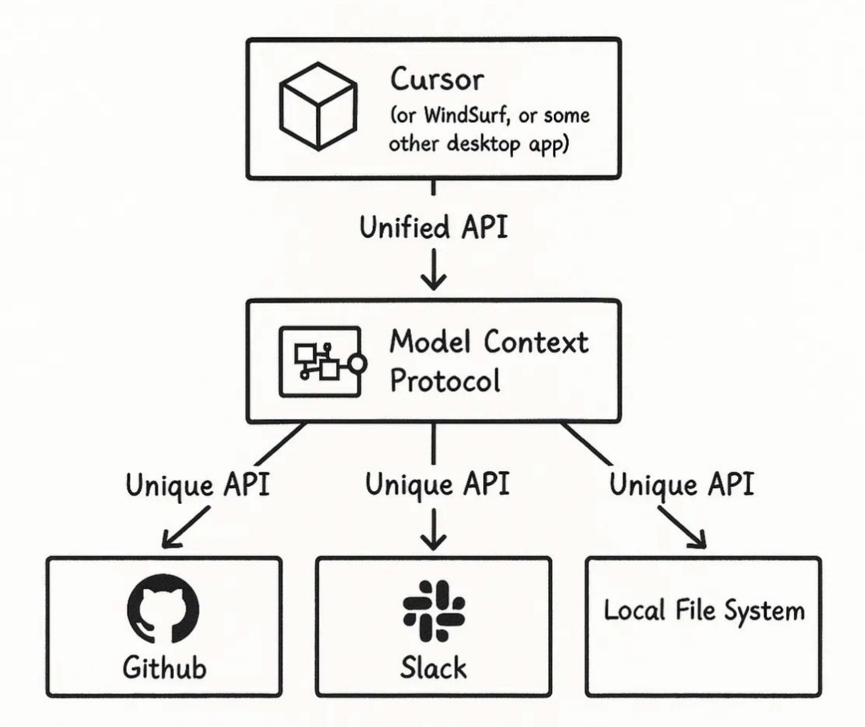
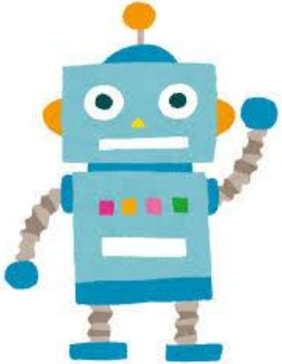


Image source

What’s an agent skill



MCP
Hammer

Skill
Manual + 🛠️

Closer to “Instead of checking intermediate steps, check only the final product”

Where have I used it

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#Gemini3HackathonTokyo was fun.
Used a bunch of Gemini APIs, learned about audio video manipulation.
Need to catch up on claude skills.

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MistralAI Worldwide Hackathon Tokyo was great !
Did not expect to pull an all-nighter, ML hackathons are intense. Used hugging face, weights and biases for the first time. Fine-tuning has changed a lot since my uni days 👴

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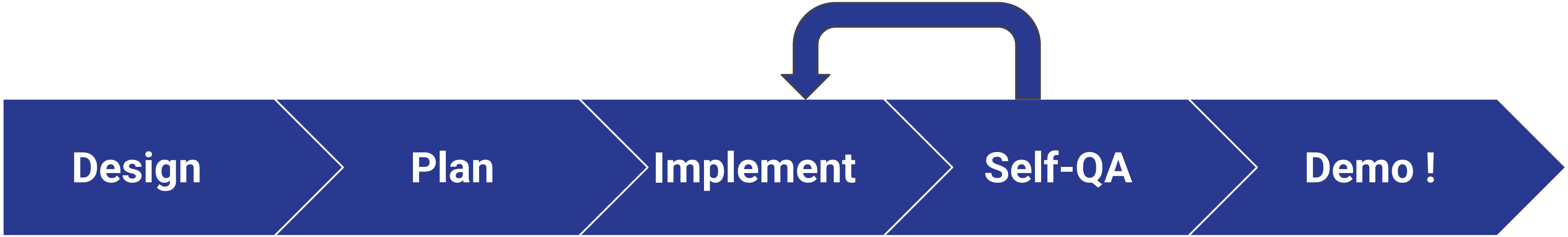
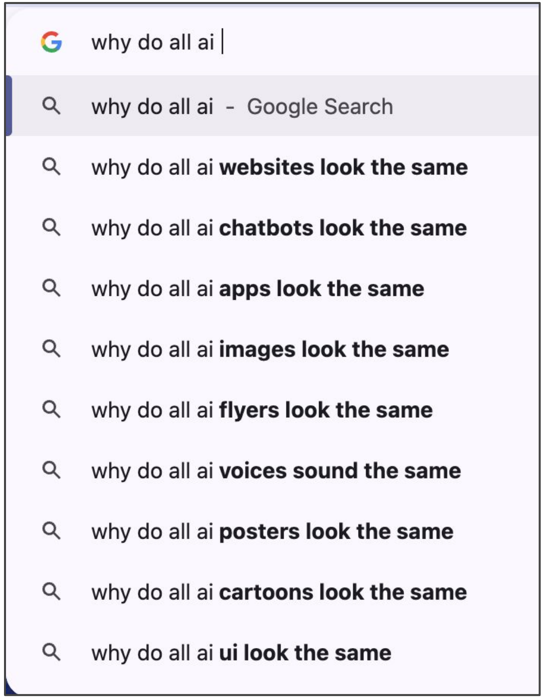
Progress



1. Design: It’s not just a phase, mom

Purple, purple everywhere



1. Design: It’s not just a phase, mom

Why Purple, purple everywhere?

- Guides often refer this palette
- Tailwind becomes golden hammer
- Tailwind starts to lose clients

BUSINESS INSIDER

TECH

Tailwind lays off 75% of its 4-person engineering team, citing 'brutal impact AI has had on our business'

By Henry Chandonnet + Follow

Tailwind's CEO said the startup's revenue has decreased by 80% amid the rise of AI. id-work/Getty Images

1. Design skills

- frontend-design
- impeccable

1. Design skills

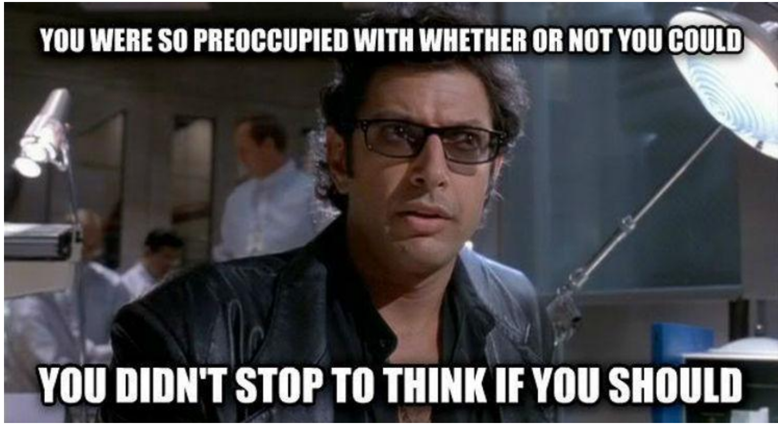
- [frontend-design]
- [impeccable]
- <many more>
    - There are several skills to get the job done
    - This talk suggests a few skills,
    - And outlines an overall approach.

1. Design skills

- Identify a “theme”
    - Fintech: Typically blue green to instill trust
    - SaaS
    - Corporate friendly, or edgy
- Root colors - primary, secondary, accent
- Enforce colors via AGENTS.md

2. Plan with `the-fool`

- Agents are good for “how”, but not for “why”
  - How this feature vs why this feature
- The discussion around “why” helps identify blind spots, and whether we should build the feature in the first place.



2. Plan with `the-fool`

4 modes

- Question assumptions
- Build counter-arguments
- Find weaknesses
- You choose (semi-autonomous)

- Relies heavily on `AskUserQuestion`
- Socratic thinking

3. Implement

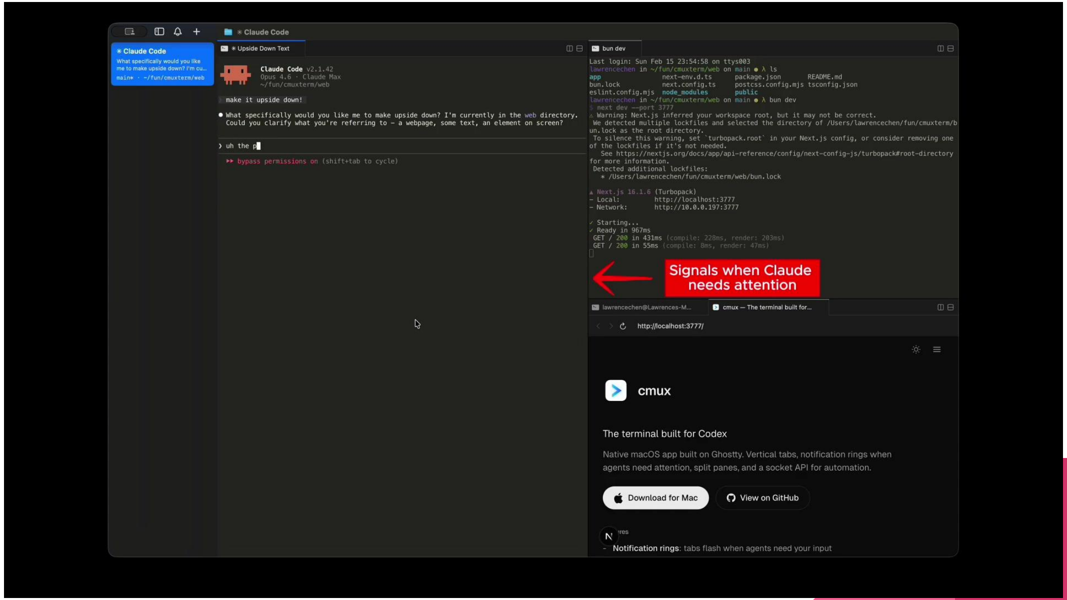
Several options here for orchestration, spinning up sub-agents

- obra/superpowers
- Oh my claude
- Claude’s own team of agents

3. Implement

Terminals that aid agentic programming workflows

- [Cmux]
- [peon-ping]



Others

- [Superset]
- [Air.dev]

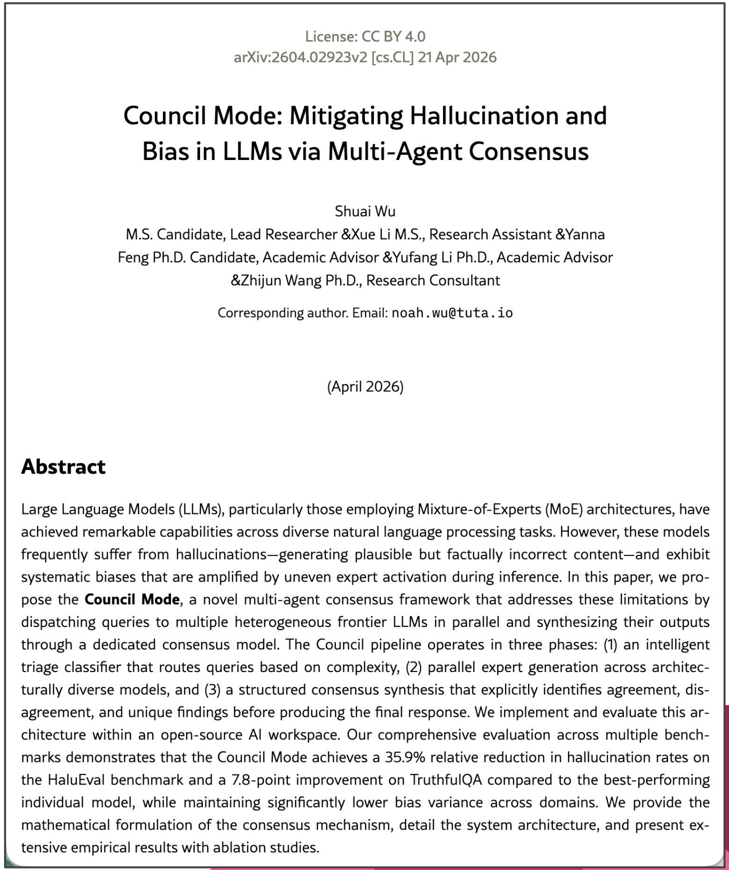
3. Implement

Language specific skills

- Python-pro
- React-expert
- Fullstack-guardian : for integration tasks
- Go find skills for your language and task

4. QA

- Largely ~~human-driven~~ human-bottlenecked
- Can use team of agents to validate here
  - arXiv
  - MoE (Mixture of Experts)
    - Need to explore this myself



arXiv:2604.02923v2 [cs.CL] 21 Apr 2026

Council Mode: Mitigating Hallucination and Bias in LLMs via Multi-Agent Consensus

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(April 2026)

Abstract

Large Language Models (LLMs), particularly those employing Mixture-of-Experts (MoE) architectures, have achieved remarkable capabilities across diverse natural language processing tasks. However, these models frequently suffer from hallucinations—generating plausible but factually incorrect content—and exhibit systematic biases that are amplified by uneven expert activation during inference. In this paper, we propose the **Council Mode**, a novel multi-agent consensus framework that addresses these limitations by dispatching queries to multiple heterogeneous frontier LLMs in parallel and synthesizing their outputs through a dedicated consensus model. The Council pipeline operates in three phases: (1) an intelligent triage classifier that routes queries based on complexity, (2) parallel expert generation across architecturally diverse models, and (3) a structured consensus synthesis that explicitly identifies agreement, disagreement, and unique findings before producing the final response. We implement and evaluate this architecture within an open-source AI workspace. Our comprehensive evaluation across multiple benchmarks demonstrates that the Council Mode achieves a 35.9% relative reduction in hallucination rates on the HaluEval benchmark and a 7.8-point improvement on TruthfulQA compared to the best-performing individual model, while maintaining significantly lower bias variance across domains. We provide the mathematical formulation of the consensus mechanism, detail the system architecture, and present extensive empirical results with ablation studies.

5. Demo 🎉

Utilize npm libraries (like remotion) that make videos programmatically

Layered with skills for it: remotion-best-practices

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🤹 Blueprint of an agent skill

🙋‍♂️

SKILL.md*

Required

🎩

📚 references
base memory

🥎

📝 assets
templates

🖐️

🏃 scripts

(naming convention)

Skill.md

SKILL.md .../the-fool | report-memory-management.md 9+, U

agent-skills-fullstack-development > .agents > skills > the-fool > SKILL.md > # The Fool

Kaustubh Hiware, last month | 1 author (Kaustubh Hiware)

```
---
name: the-fool
description: Use when challenging ideas, plans, decisions, or proposals using structured critical reasoning. Invoke to play devil's advocate, run a pre-mortem, red team, or audit evidence and assumptions.
license: MIT
metadata:
  author: https://github.com/Jeffallan
  version: "1.0.0"
  domain: workflow
  triggers: play the fool, devil's advocate, challenge this, stress test, poke holes, what could go wrong, red team, pre-mortem, test my assumptions
  role: expert
  scope: review
  output-format: report
  related-skills: architecture-designer, code-reviewer, feature-forge
---

# The Fool

The court jester who alone could speak truth to the king. Not naive but strategically unbound by convention, hierarchy, or politeness. Applies
```

metadata

What's in the system prompt

CLAUDE.md

ALL OF IT

All Skills

Metadata only

When should a rule be a skill ?

Depends (the diplomatic answer)

Depends on what task the agent serves in a folder

- Writing code
- Reviewing code
- Operations (db, migrations)
- Understanding new codebase (for new contributors)

When should a rule be a skill ?

Programming principles are only relevant when making code changes.

So, only when editing files.

Skills are “lazy loaded”, and can be triggered specifically for agent edit operations.

TL;DR: All my repos have coding principles as a skill, not rules.

Agent spec should be at most 200 lines anyway.

References for making skills

- Agent skills homepage
- Anthropic's Skills Course
    - /skill-creator → A skill that creates skills 🔄
- The-Complete-Guide-to-Building-Skill-for-Claude.pdf

Plugins & Marketplaces

Layers of abstractions

- Skill: Local directory
- Plugin: Directory in the cloud that you can install and update (like packages)
- Marketplace: Places to look plugins for

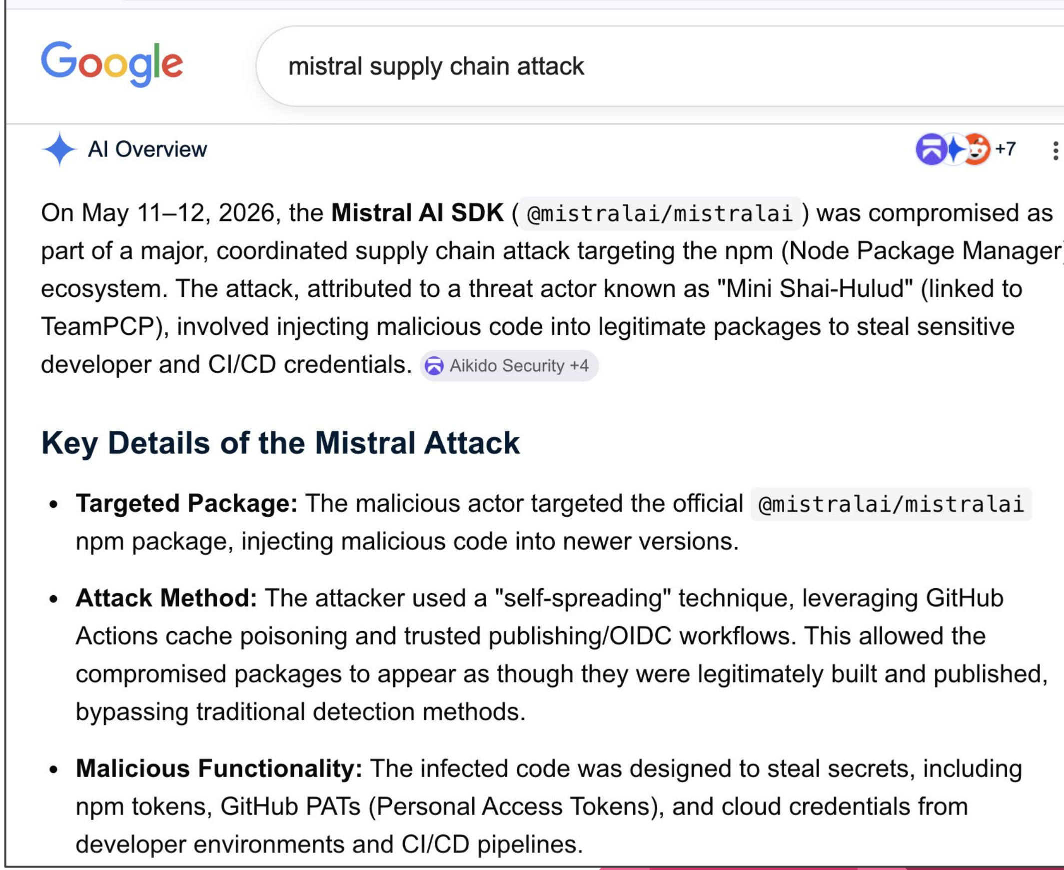
🖐 About using plugins in corporate

Supply chain attacks

[LLM03:2025 Supply Chain - OWASP Gen AI Security Project]

- When your project becomes vulnerable due to a downstream dependency

Today Mistral had a similar attack (he said, wearing a Mistral shirt)



Google

mistral supply chain attack

AI Overview

On May 11–12, 2026, the **Mistral AI SDK** (`@mistralai/mistralai`) was compromised as part of a major, coordinated supply chain attack targeting the npm (Node Package Manager ecosystem. The attack, attributed to a threat actor known as "Mini Shai-Hulud" (linked to TeamPCP), involved injecting malicious code into legitimate packages to steal sensitive developer and CI/CD credentials. Aikido Security +4

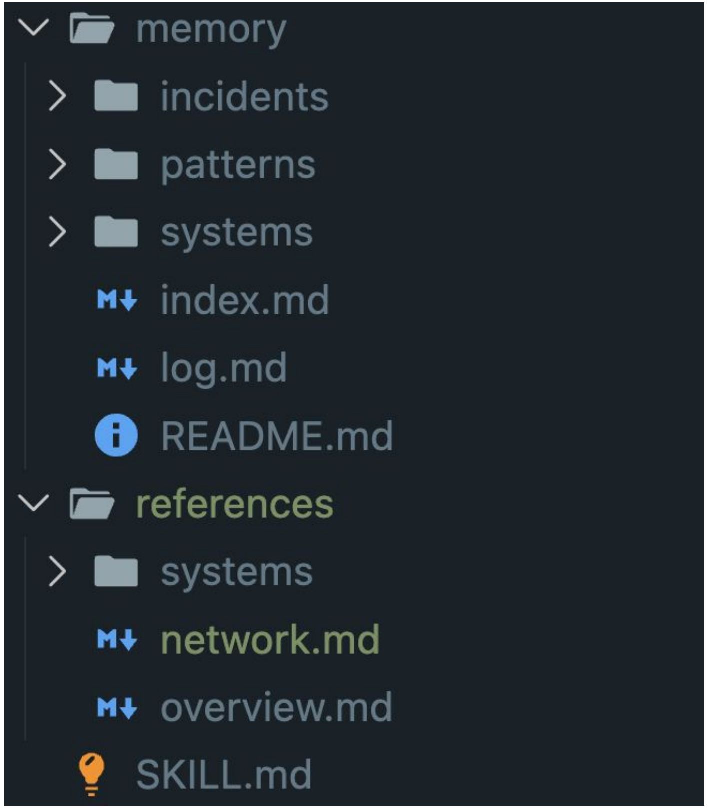
Key Details of the Mistral Attack

- **Targeted Package:** The malicious actor targeted the official `@mistralai/mistralai` npm package, injecting malicious code into newer versions.
- **Attack Method:** The attacker used a "self-spreading" technique, leveraging GitHub Actions cache poisoning and trusted publishing/OIDC workflows. This allowed the compromised packages to appear as though they were legitimately built and published, bypassing traditional detection methods.
- **Malicious Functionality:** The infected code was designed to steal secrets, including npm tokens, GitHub PATs (Personal Access Tokens), and cloud credentials from developer environments and CI/CD pipelines.

Memory: Progressive overload ✨

- \`references\` contains the curated domain knowledge relevant for agents.
  - *Referenced* in SKILLS.md, only read when relevant.
  - Read at runtime
- \`scripts\` - useful to execute clis
  - K8S, docker status check
  - Generate and execute a SQL query
- \`assets\` / \`templates\`
  - Used by the others

Memory: Progressive overload ✨



memory makes most of the initial knowledge base.

For a long running agent, “logs” will probably be biggest (next slide)

The 3 types of Memory

1. Tribal knowledge
   a. Existing practices, know-how of tools required, where to look, how to do things
   b. No WRITE, only READ
2. Execution log
   a. Whenever the agent is invoked, for instance, handling inquiries / incident response
   b. Running log of operations
   c. Heavy WRITE
3. Patterns
   a. Consolidated compendium
   b. Refer during problem solving
   c. Light WRITE: When solutions are found, for future incidents

Updating memory

1. Tribal knowledge
    a. Human edits only
2. Execution log
    a. Enforce agent to populate this with template file `YYYY_MM_DD_OPERATION.md`
    b. Do not edit existing logs
3. Patterns
    a. Only write upon task completion (Can add a user confirmation step)
    b. Before adding a pattern: Check for overlap with existing patterns
    c. Only add new pattern if none match uniquely

Memory optimization

There are a couple of skills around this, need to explore them myself.

- Mempalace
- Caveman

Reportedly reduces token usage (oh, those precious tokens !)

Outline

- What’s an agent skill
- Skills for full-stack development
- Making a skill and giving it memory

▶ **Token observability** 🍰

Ghostty File Edit View Window Help

Sat, 14 Mar 100% kaustubh 1:04 AM

accounting: ready

```
Claude Code v2.1.74

        Welcome back K!                 Tips for getting started
                                        Run /init to create a CLAUDE.md file with instructions for Claude

                                        Recent activity
                                        No recent activity
Sonnet 4.6 · Claude Pro ·
                 's Organization
  ~/bytes_n_pieces/accounting

❯ s

accounting  Model: Sonnet 4.6  ⌥ main  5h: 40%  7d: 0%                         In colored-usage.sh

        Welcome back K!                 Run /init to create a CLAUDE.md file with instructions for Claude

                                        Recent activity
                                        No recent activity
Sonnet 4.6 · Claude Pro ·
                 's Organization
     ~/GitHub/dotfiles

❯

dotfiles  Model: Sonnet 4.6  ⌥ master  5h: 40%  7d: 0%

        Welcome back K!                 Run /init to create a CLAUDE.md file with instructions for Claude

                                        Recent activity
                                        No recent activity
Sonnet 4.6 · Claude Pro ·
                 's Organization
     ~/GitHub/NotiFyre

❯

NotiFyre  Model: Sonnet 4.6  ⌥ master  5h: 40%  7d: 0%
```

Tokens observability 🍰

- Ccstatusline
    - What I use: Track context, 5h window, week token consumption
    - Extremely configurable
- Omc-hud
    - Similar, can also track active agents, which agents are awaiting approval
- Claude-pace
- cclogviewer

Looking under the hood

- [Cclogviewer]

Claude starting to hide its

intermediate operations

Assistant 08:57:28 Conversation size: 15,256 | [+]

Hello! How can I help you today?

User 09:35:08 ~7 tokens

Please read brad-cc-hours.txt

Assistant 09:35:14 Conversation size: 15,271 | [+]

› Read /Users/brad/Desktop/brad-cc-hours.txt toolu_01KSexgs9vj8efA38yyrhXfv

Assistant 09:35:23 Conversation size: 17,190 | [+]

The file shows your Claude usage analysis over 5 weeks (weeks 27-31 of 2025). Here are the key highlights:

Weekly Activity:
- Week 27: 80h active, 29.9h conversations
- Week 28: 96h active, 48.3h conversations
- Week 29: 86h active, 46.2h conversations
- Week 30: 84h active, 34.3h conversations
- Week 31: 28h active, 16.0h conversations (partial week)

Overall Summary:
- Average 74.8h of activity per week
- Average 34.9h of active conversations per week
- 1.32x parallel session factor

Model Usage:
- Opus 4: 22.4h/week average (within 24-40h limit)
- Sonnet 4: 21.2h/week average (within 240-480h limit)
- Total parallel sessions: 46.1h/week

Summary

- What’s inside a skill
- How to manage memory
- Few recommended skills for full stack development

Make your agent more skillful !

🤹 Agent Skills for full-stack development

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2026.05.12